

Akila Abenayaka

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[GitHub](#) | [LinkedIn](#) | [Medium](#) | [Portfolio](#)

SOFTWARE ENGINEERING UNDERGRADUATE

Passionate Software Engineering undergraduate at the University of Kelaniya with strong interests in **Generative AI**, **Full-Stack Development**, and **Cloud-Native Systems**. Experienced in developing scalable web applications using MERN and Spring Boot ecosystems while exploring AI-powered solutions using LLMs and RAG architectures. A fast learner and collaborative team player committed to building impactful, intelligent software solutions.

TECHNICAL SKILLS

Languages	: Java, JavaScript, TypeScript, Python, C, HTML, CSS, SQL
Frontend	: React.js, Next.js, Redux Toolkit, Tailwind CSS, Bootstrap
Backend	: Node.js, Express.js, Spring Boot, REST APIs
AI & Data	: Generative AI, Gemini API, RAG Architecture, Prompt Engineering
Databases	: PostgreSQL, MongoDB, MySQL, Redis
Tools & Platforms	: Git, Docker, Firebase, Supabase, Postman, Vercel

EDUCATION

BSc (Hons) Software Engineering <i>University of Kelaniya</i>	2024 – Present Sri Lanka
- Specializing in Net-Centric Web Application Development, Data Science & Engineering, and Mobile Computing - CGPA: 3.729 / 4.0	
GCE Advanced Level <i>Kegalu Vidyalaya</i>	2007 – 2020 Kegalle, Sri Lanka
- Completed coursework in Combined Mathematics, Physics, and Chemistry 2A & 1C Passes Z-Score: 1.589	

PROJECTS

AI-Driven Intelligent Chatbot <i>Individual Project</i>	<i>Python, Gemini 2.5 Pro, PostgreSQL</i>	GitHub
- Developed an AI-powered chatbot utilizing Gemini 2.5 Pro and Retrieval-Augmented Generation (RAG) for context-aware responses - Built a custom knowledge retrieval system using PostgreSQL for efficient document querying and low-latency responses - Integrated vector search and PDF querying capabilities to enhance external knowledge access - Optimized prompt engineering techniques to improve conversational quality and response accuracy		
Velora E-Commerce Platform <i>Individual Project</i>	<i>MERN Stack, Redux Toolkit, Stripe</i>	Live Demo
- Engineered a scalable full-stack clothing marketplace with secure authentication and responsive UI design - Managed application-wide state using Redux Toolkit to ensure seamless frontend data flow - Integrated Stripe for secure payment processing and Cloudinary for media management - Implemented RESTful APIs and optimized frontend responsiveness across mobile and desktop devices		

- Architected a scalable, decoupled booking platform utilizing a **Microservices Architecture** with Spring Boot and PostgreSQL data isolation per service.
- Implemented asynchronous inter-service communication and event-driven workflows using **RabbitMQ** to handle real-time booking and notification pipelines.
- Integrated **Asgardeo** for robust Identity and Access Management (IAM), implementing secure OAuth2/OIDC protocols for user authentication and role-based authorization.
- Leveraged **Netflix Eureka Service Registry** for dynamic service discovery, ensuring high availability, load balancing, and fault tolerance across the ecosystem.
- Containerized the entire multi-service ecosystem using **Docker** and orchestrated multi-container deployments with **Docker Compose** for streamlined cloud deployment and local replication.

- Contributed to the official TEDx UOK event website with emphasis on UI/UX and mobile responsiveness
- Integrated **Supabase** services for secure event registration and real-time backend functionality
- Improved website SEO and optimized performance for faster asset loading and user engagement

CERTIFICATIONS & ACHIEVEMENTS

- **First Place** – Paramount Mathz IQ Competition, Department of Mathematics, University of Kelaniya
- **Generative AI Fundamentals** – Google Cloud Training (2024)
- **Participant** – **IEEEExtreme 19.0 Programming Competition** (2025)

VOLUNTEER EXPERIENCE & LEADERSHIP

Head of Logistics Software Engineering Students' Association (SESA)	2025 – Present University of Kelaniya
• Coordinating logistics and operational planning for technical events and workshops • Managing resource allocation and vendor coordination for student activities	
Digital Marketing Lead IEEE Student Branch	2024 – 2026 University of Kelaniya
• Leading digital outreach campaigns and social media engagement for IEEE events • Designed promotional content and improved audience reach for technical webinars	

LANGUAGES

English	: Professional Proficiency
Sinhala	: Native Proficiency

REFERENCES

- **Prof. (Mrs) I.U. Hewapathirana**
Head, Software Engineering Teaching Unit, University of Kelaniya
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- **Dr. Nalin Warnajith**
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